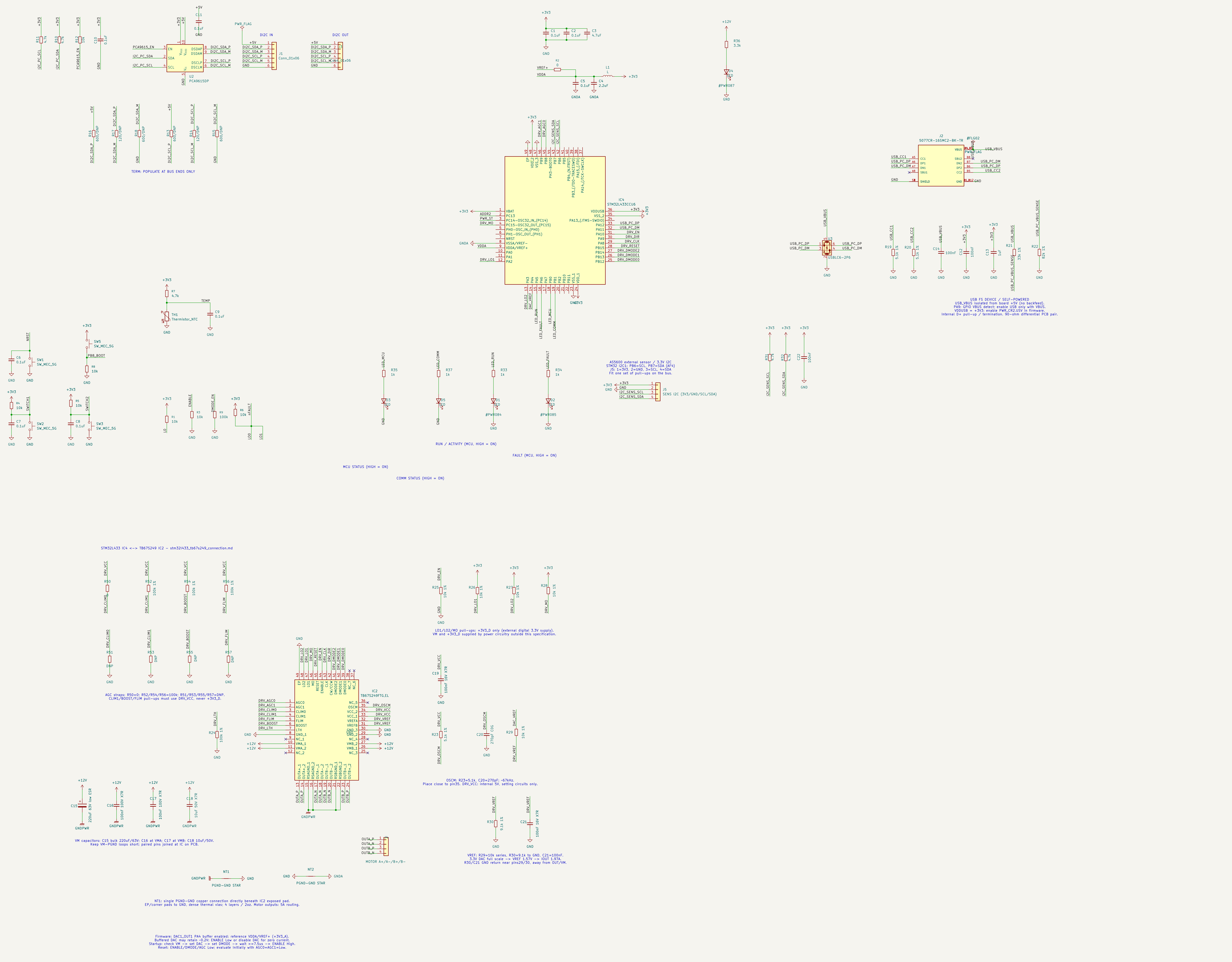

TERM: POPULATE AT BUS ENDS ONLY

USB FS DEVICE / SELF–POWERED
USB_VBUS isolated from board +5V (no backfeed).
PA9: GPIO VBUS detect; enable USB only with VBUS.
VDDUSB = +3V3; enable PWR_CR2.USV in firmware.
Internal D+ pull–up / termination. 90–ohm differential PCB pair.

AS5600 external sensor / 3.3V I2C
STM32 I2C1: PB6=SCL, PB7=SDA (AF4)
J5: 1=3V3, 2=GND, 3=SCL, 4=SDA
Fit one set of pull–ups on the bus.

RUN / ACTIVITY (MCU, HIGH = ON)

FAULT (MCU, HIGH = ON)

MCU STATUS (HIGH = ON)

COMM STATUS (HIGH = ON)

STM32L433 IC4 <–> TB67S249 IC2 – stm32l433_tb67s249_connection.md

AGC straps: R50=0; R52/R54/R56=100k; R51/R53/R55/R57=DNP.
CLIM1/BOOST/FLIM pull–ups must use DRV_VCC, never +3V3_D.

LO1/LO2/MO pull–ups: +3V3_D only (external digital 3.3V supply).
VM and +3V3_D supplied by power circuitry outside this specification.

OSCM: R23=5.1k, C20=270pF; ~67kHz.
Place close to pin35. DRV_VCC: internal 5V, setting circuits only.

VM capacitors: C15 bulk 220uF/63V; C16 at VMA; C17 at VMB; C18 10uF/50V.
Keep VM–PGND loops short; paired pins joined at IC on PCB.

VREF: R29=10k series, R30=9.1k to GND, C21=100nF.
3.3V DAC full scale –> VREF 1.57V –> IOUT 1.97A.
R30/C21 GND return near pins29/30, away from OUT/VM.

NT1: single PGND–GND copper connection directly beneath IC2 exposed pad.
EP/corner pads to GND, dense thermal vias; 4 layers / 2oz. Motor outputs: 5A routing.

Firmware: DAC1_OUT1 PA4 buffer enabled; reference VDDA/VREF+ (+3V3_A).
Buffered DAC may retain ~0.2V: ENABLE Low or disable DAC for zero current.
Startup: check VM –> set DAC –> set DMODE –> wait >=7.5us –> ENABLE High.
Reset: ENABLE/DMODE/AGC Low; evaluate initially with AGC0=AGC1=Low.

☐ U1

☐ U4

Sheet: /
File: schematic.kicad_sch

**Title:**

Size: A1
Date:
**Rev:**
KiCad E.D.A. 10.0.4
Id: 1/1